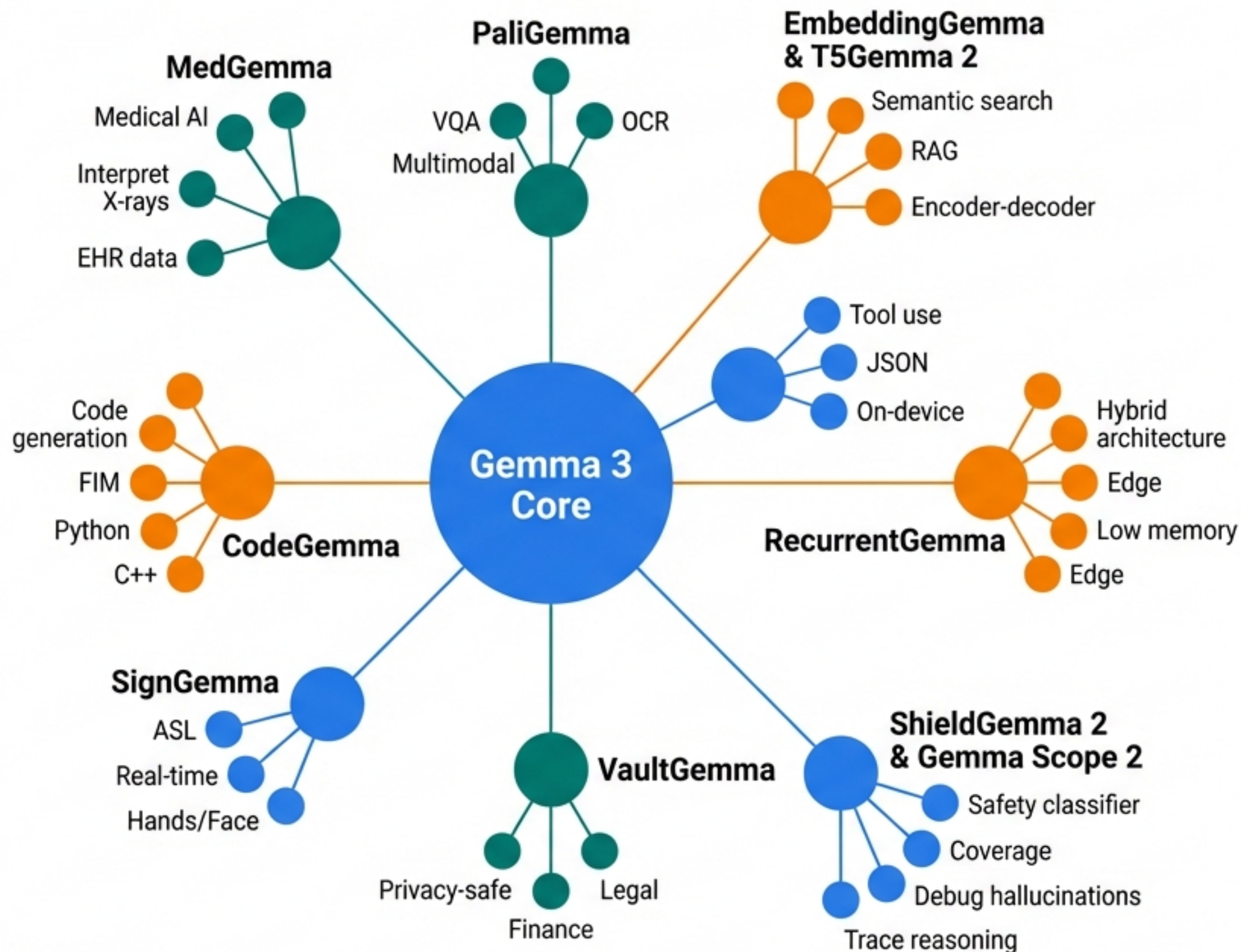


The Gemma Ecosystem

Architectures, Capabilities,
and Selection Strategies



A Technical Guide to the 2026 Open Model Family

January 2026 Edition | Google DeepMind


200M+
Downloads

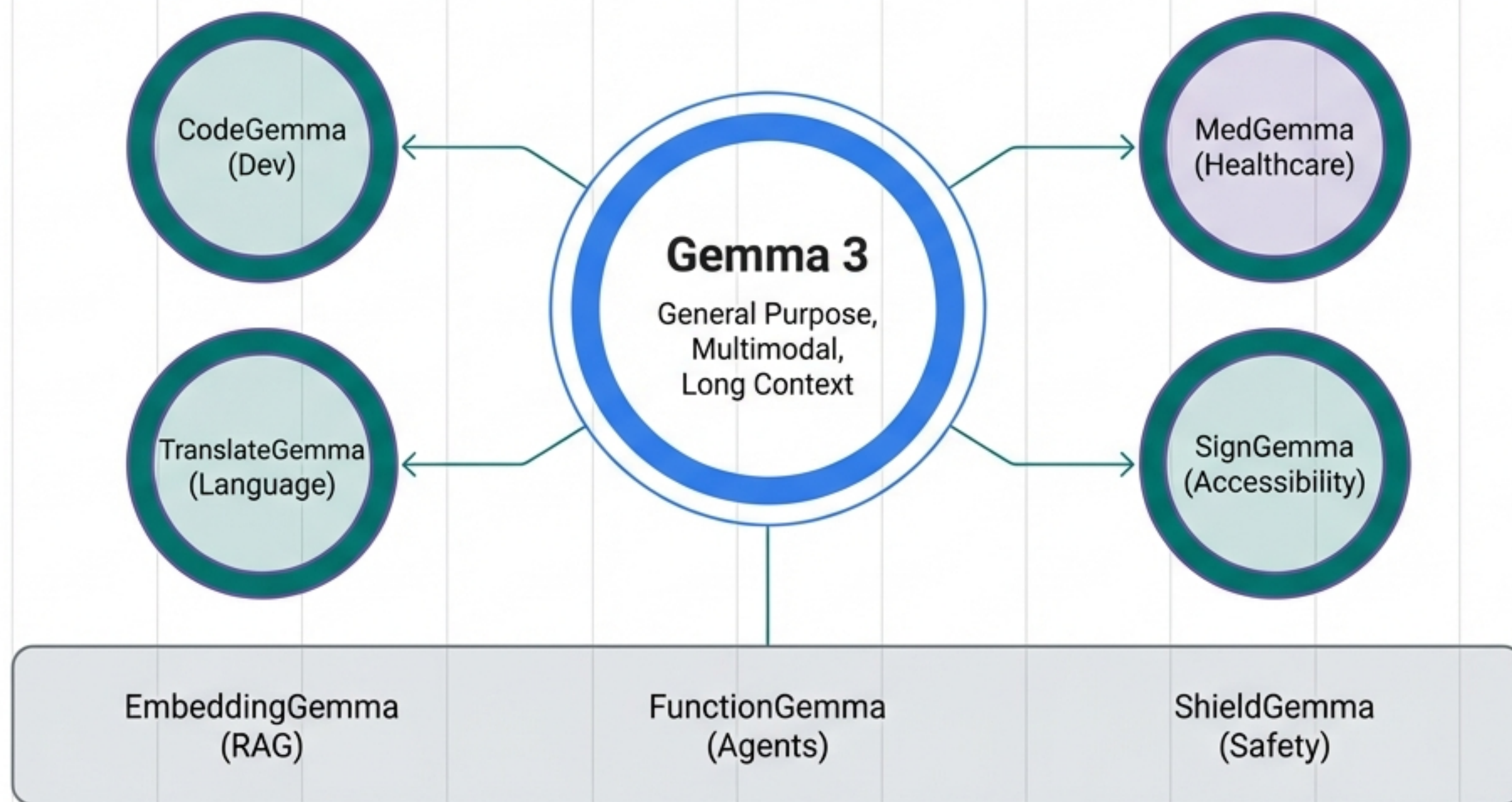

60,000+
Community Variants


140+
Languages Supported

Moving Beyond Generalists to a Specialized Ecosystem

The Hub and Spoke Architecture Strategy

The era of “one size fits all” is over. The 2026 Gemma architecture enables developers to route tasks based on utility, balancing raw reasoning power with edge efficiency.



Efficiency-First Routing

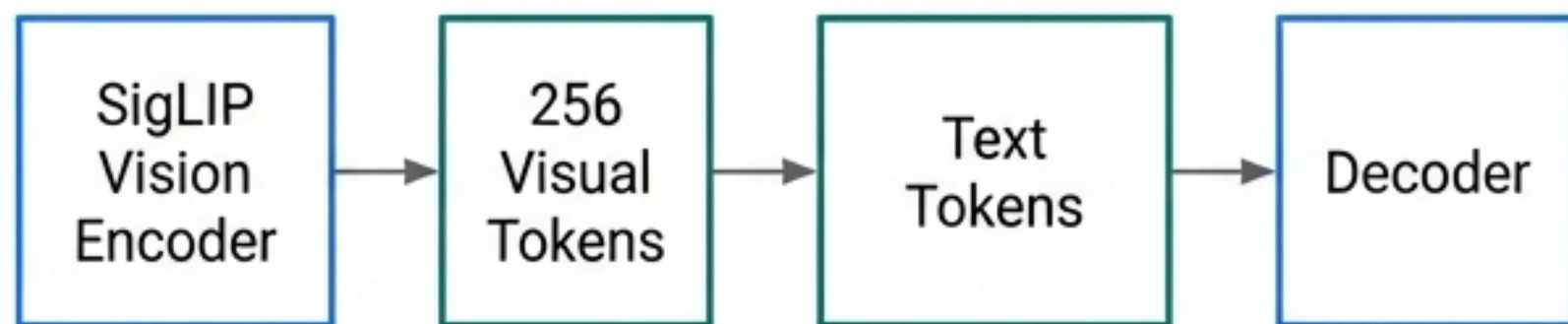
A 270M parameter model handles routing, while a 27B model activates only for complex reasoning.

The Core Intelligence: Gemma 3

Model Specifications

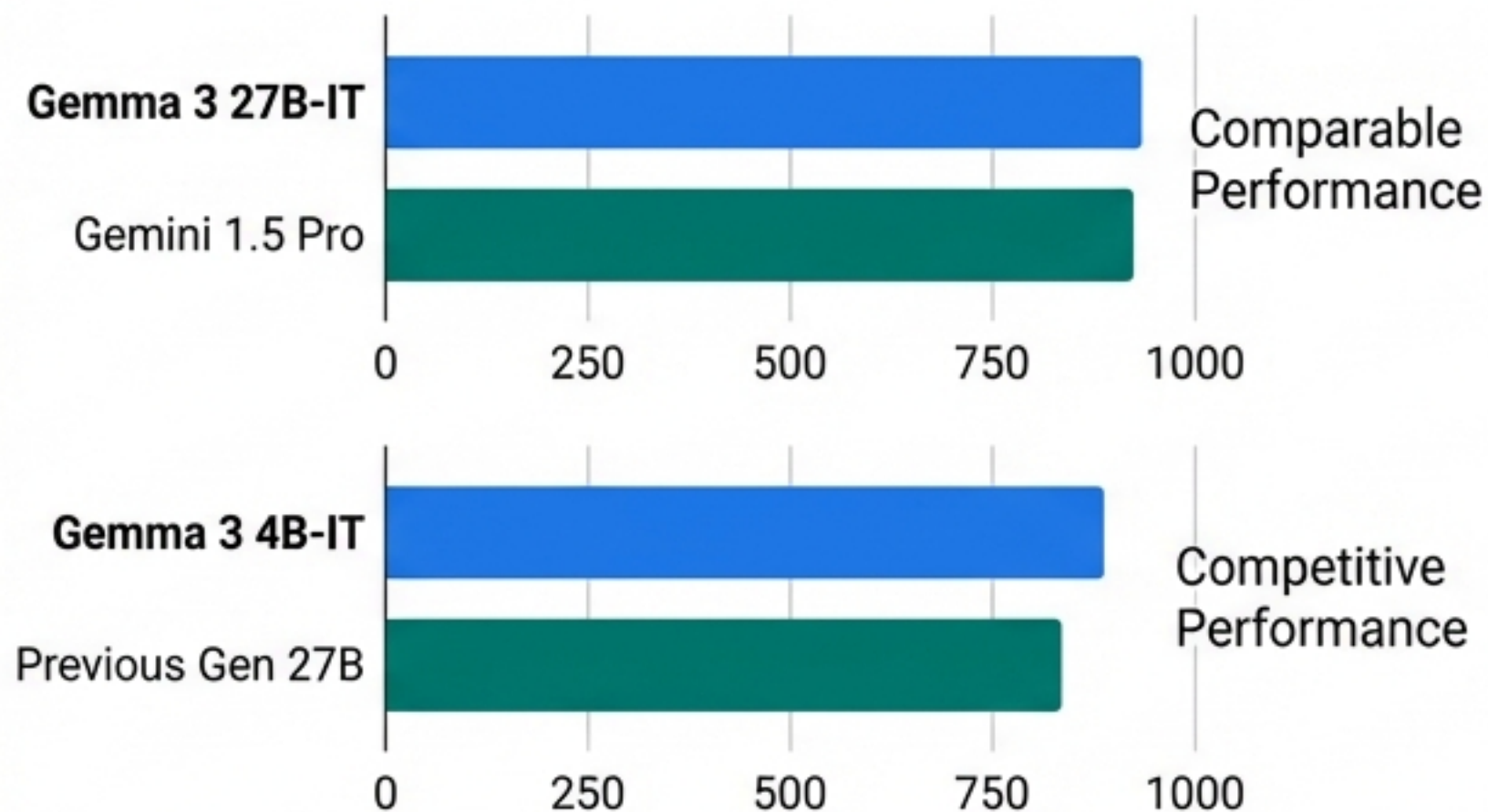
- **Release:** March 2025
- **Parameters:** 1B, 4B, 12B, 27B
- **Context Window:** 128K tokens (4B-27B), 32K (1B)
- **Architecture:** Decoder-only transformer, Grouped-Query Attention (GQA)

Multimodal Architecture



Pan & Scan algorithm for adaptive cropping up to 896x896 resolution

Performance Benchmarks



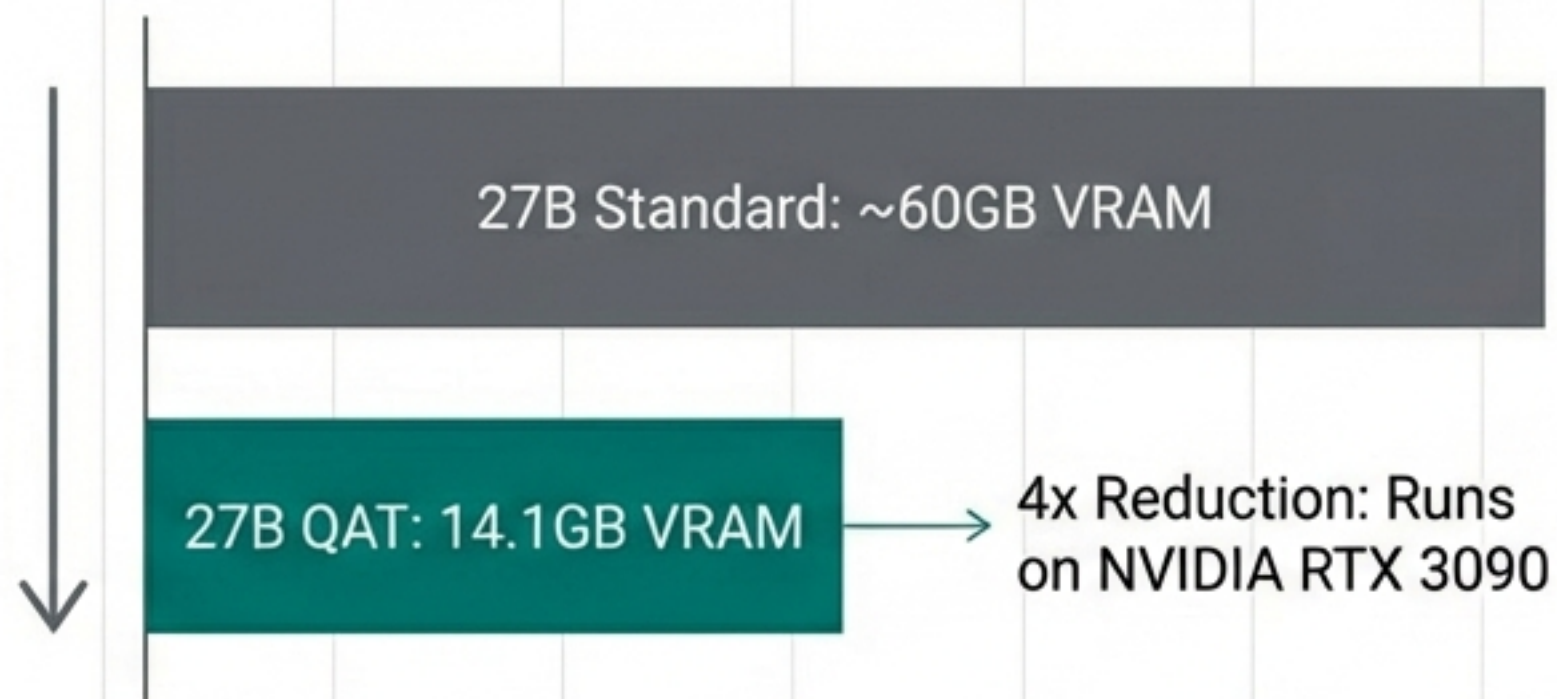
Attention Mechanism

Sliding Window Attention (5 local : 1 global) ensures efficient long-context processing.

Hyper-Efficiency via QAT and Micro-Models

Gemma 3 QAT (Quantization-Aware Training)

Simulating quantization during training (~5,000 steps) prevents the quality degradation seen in post-training quantization.



Gemma 3 270M (The 'Micro' Model)

A 270M parameter model (100M transformer + 170M embeddings) for router/classifier tasks.



0.75% of Pixel 9 Pro battery
consumed after 25 full conversations.

Mobile-First Architecture with Gemma 3n

E4B Model (~4B Params)

E2B Model (~1.91B Params)

MatFormer: Shared Parameters

Dynamic parameter activation
based on device constraints.

Multimodal Inputs



MobileNet-V5 (Vision) + USM
Audio Encoder (6.25 tokens/sec).

Video Processing

Supports 60 FPS video
input processing directly
on Pixel devices.

Memory Optimization

Per-Layer Embedding (PLE)
Caching and KV Cache
Sharing (2x speedup).

Advanced Vision-Language Transfer with PaliGemma 2

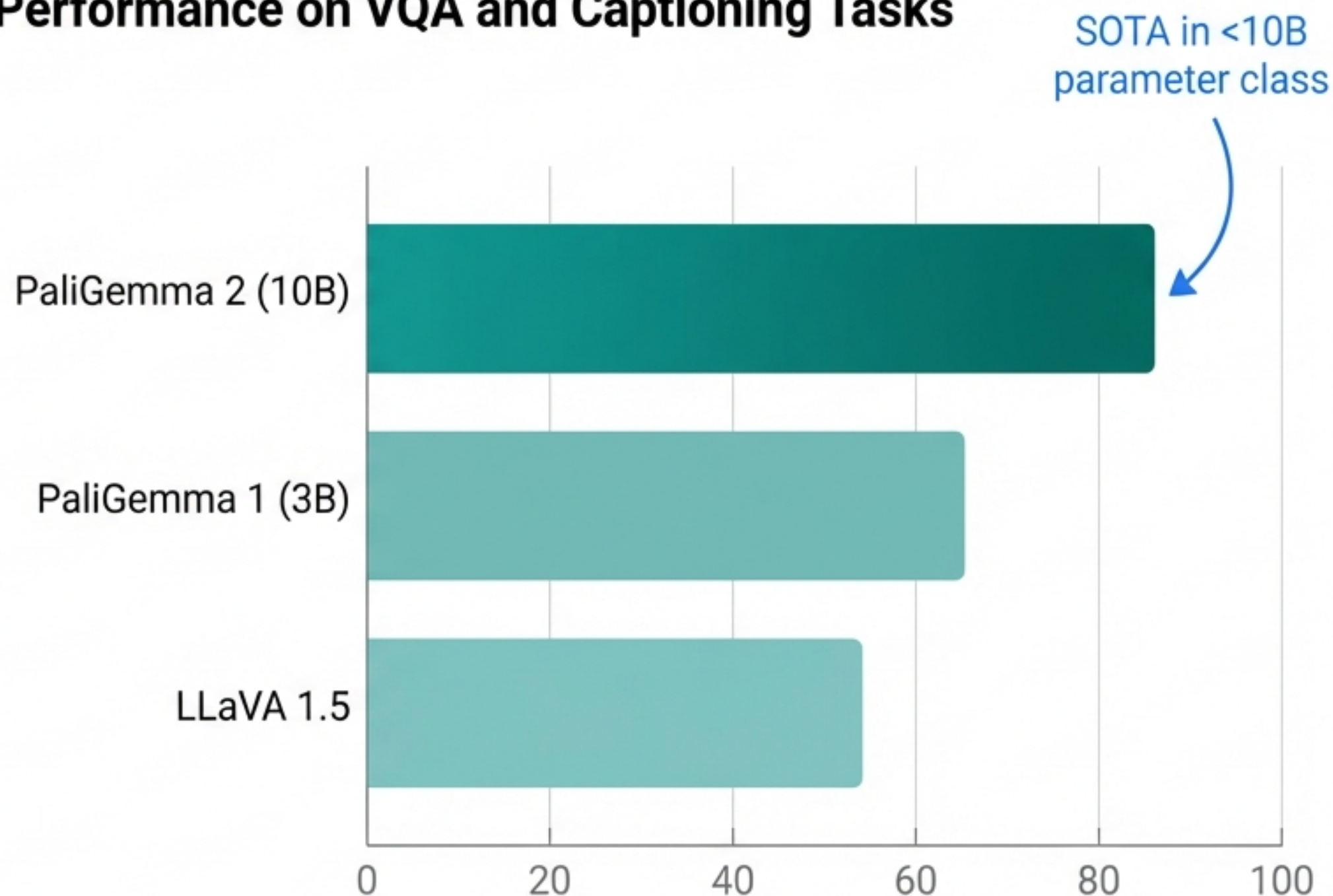
Architecture: SigLIP-So400m vision encoder + Gemma 2 decoder

- Sizes: 3B, 10B, 28B
- Resolutions: 224px, 448px, 896px

Key Capabilities:

- **OCR:** Fine-tuned for dense text, molecular structures, and music scores.
- **Segmentation:** Generates bounding boxes and masks.
- **Doc Understanding:** SOTA on ChartQA and DocVQA.

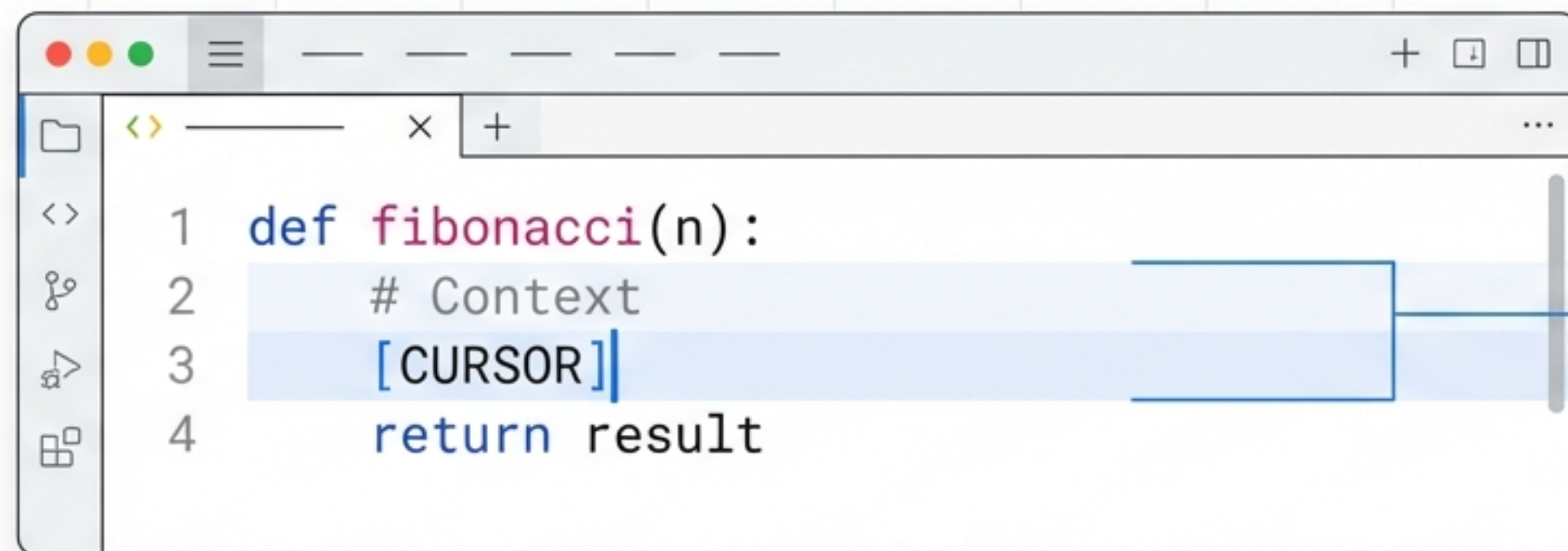
Performance on VQA and Captioning Tasks



Accelerating Development with CodeGemma

Model Variants

- **7B / 7B-IT**: Code completion, debugging, chat
- **2B**: Latency-optimized for local IDE auto-complete



```
1 def fibonacci(n):  
2     # Context  
3     [CURSOR]  
4     return result
```

Fill-In-The-Middle (FIM)
Google Blue (#1A73E8)

Specs & Performance

- Context: 8K tokens
- Training Data: 500B tokens (80% code, 20% natural language)
- Languages: Python, Java, Rust, C++, Go, Kotlin, JS
- Benchmark: 74%+ on HumanEval (Python), outperforming StarCoder.

Breaking Language Barriers: TranslateGemma & SignGemma

TranslateGemma

High-fidelity translation across 55 languages.

12B Model Performance > **Gemma 3 27B Baseline**

Methodology:

Trained via SFT + RL (MetricX-QE reward models).

Deployment:

4B (Mobile), 12B (Laptop), 27B (Server).

SignGemma (2025)



Real-time ASL to English Translation.

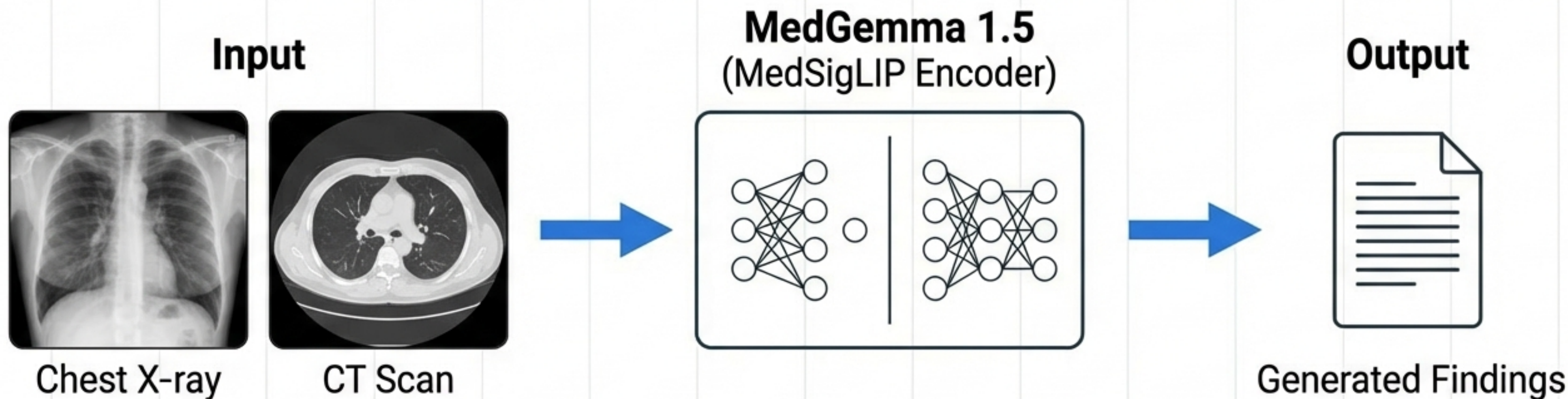
Specs:

<200ms latency, fully on-device processing.

Input:

Vision Transformer analyzes hands, body pose, and facial expression.

Medical Imaging & Reasoning with MedGemma 1.5



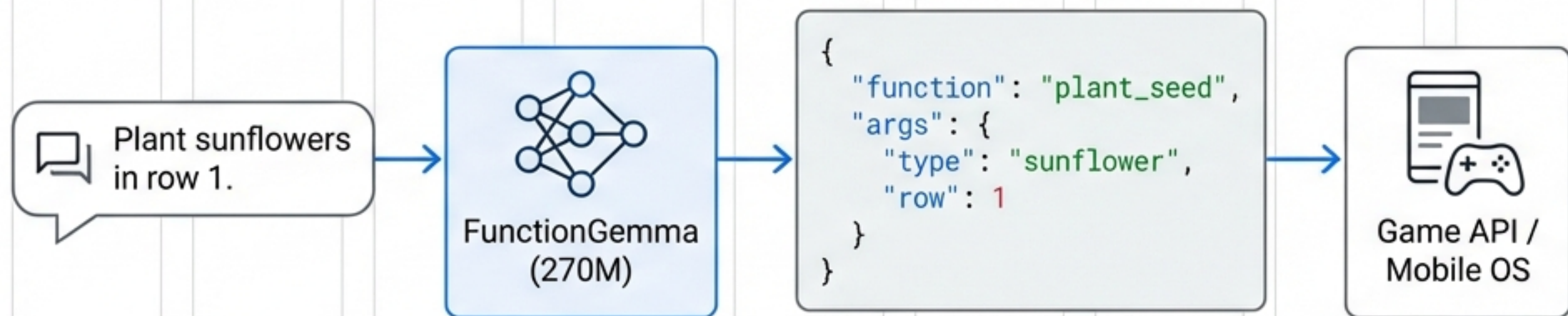
Key Features

- **Localization:** +35% IoU improvement on anatomical structures.
- **EHR Integration:** ~90% accuracy on electronic health record understanding.
- **Time-Series:** Analyzes disease progression across longitudinal data.

DISCLAIMER: Research model. Not clinical-grade. Requires human oversight.

Powering Agentic Workflows with FunctionGemma

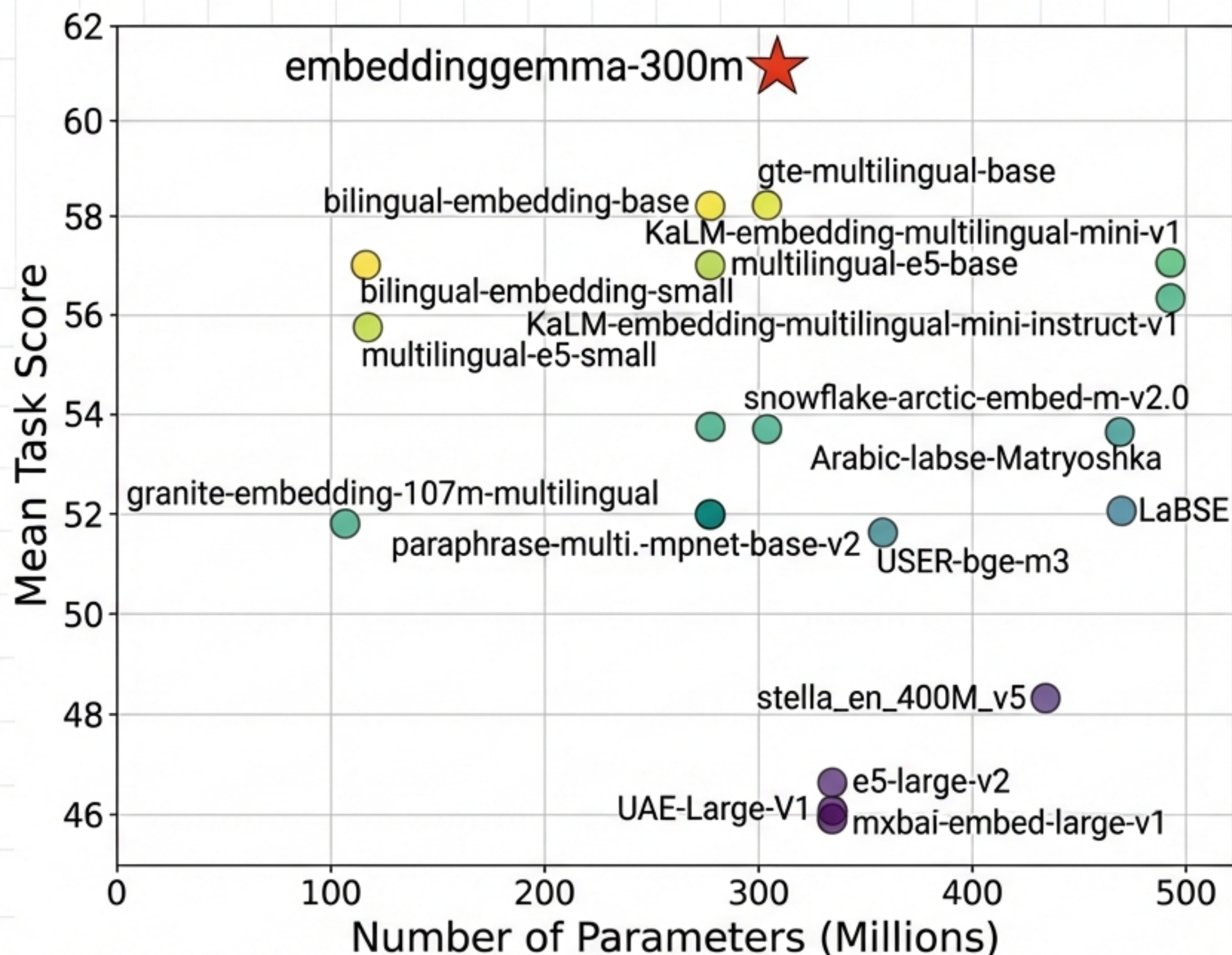
Bridging the gap between natural language and executable APIs.



Performance Metrics

- Accuracy: ~85% on "Mobile Actions" dataset (vs 58% base model).
- Speed: ~50 tokens/sec on mobile CPU.

On-Device RAG with EmbeddingGemma



Key Highlights

- **Size:** 308M Parameters
- **Feature:** Matryoshka Representation Learning (MRL)
 - Flexible output dimensions (128 to 768) for storage/speed trade-offs.
- **Benchmark:** #1 on MTEB (Multilingual, Code) for models <500M.
- **Use Case:** Private, offline Retrieval Augmented Generation (RAG).

Specialized Architectures: Privacy & Recurrence



VaultGemma (Privacy-First)

- Mechanism: Differential Privacy (DP-SGD) with $\epsilon \leq 2.0$.
- Guarantee: Zero memorization of training data (tested on 1M samples).
- Application: Medical records, financial data.



RecurrentGemma (Infinite Context)

- Architecture: “Griffin” Hybrid (Gated Linear Recurrences + Local Attention).
- Efficiency: Fixed state size. Memory usage remains constant regardless of sequence length.
- Application: Processing infinite logs, streaming data on edge devices.

Safety & Interpretability Suite



ShieldGemma 2

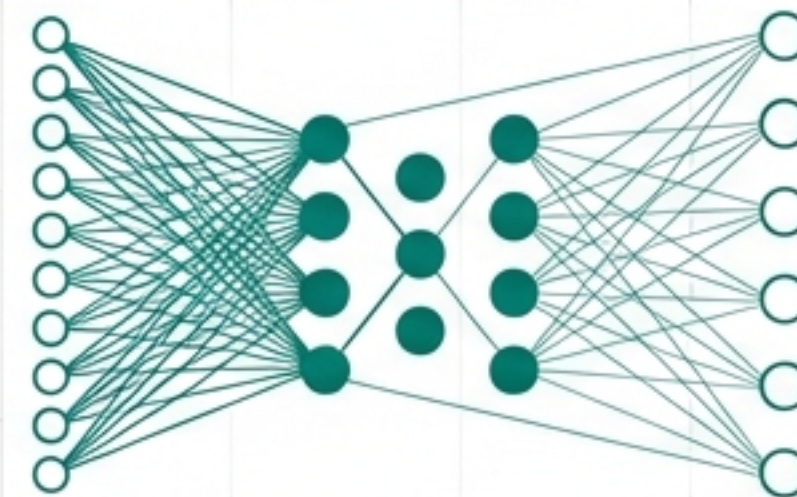
Multimodal safety classifier
(Text + Image).

- ✓ Sexually Explicit
- ✓ Dangerous Content
- ✓ Violence
- ✓ Hate Speech

Outperforms LlavaGuard and GPT-4o mini
on internal benchmarks.



Gemma Scope 2



Open Interpretability Platform.

- **Scale:** 110 Petabytes of activation data.
- **Tech:** Sparse Autoencoders (SAEs) and Transcoders.
- **Goal:** Debug hallucinations and trace reasoning at the neuron level.

Architect’s Selection Guide (Jan 2026)

Requirement	Recommended Model	Hardware Target	Key Specs
 General Reasoning	Gemma 3 (12B/27B)	Server / High-End Desktop	128K Context, Multimodal
 Mobile / Edge	Gemma 3n / 270M	Pixel 9 / iPhone / Raspberry Pi	0.75% Battery drain
 Coding Assistant	CodeGemma	Local IDE (Laptop)	FIM, 8K Context
 Vision / OCR	PaliGemma 2	GPU (20GB+)	896px Resolution
 Data Privacy	VaultGemma	Secure Enclave	DP-SGD, Epsilon < 2.0
 Agentic Actions	FunctionGemma	Mobile CPU	JSON Output, 50 tok/sec

Deployment & Ecosystem Integration



Deployment Stacks

- **Cloud:** Vertex AI (TPU optimized), AWS SageMaker
- **On-Device:** Android AI Edge, MediaPipe, MLX (Apple Silicon)
- **Local:** Ollama (Q4_0), LM Studio, Hugging Face Transformers



Scan for
Model Cards



Fine-Tuning

- Native support in Keras, JAX, and PyTorch.
- LoRA (Low-Rank Adaptation) supported across all sizes.
- Cookbook resources for RAG and Agents.



Future Roadmap (2026)

- **Gemma 4:** Native video understanding.
- **SignGemma:** Expansion to bidirectional translation.
- **MedGemma:** Entering clinical trials.



Scan for
Model Cards



Gemma
Cookbook